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MILITARY SPECIFICATION

DRIVER TESTING AND TRAINING DEVICES, PORTABLE AND SIMULATOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers driver testing and training devices, portable and simulator.

1.2 Classification. The driver testing and training devices shall be of the following types and classes, as specified (see 6.2):

Type P - Portable.
Type S - Simulator.
Class 1 - Van-mounted.
Class 2 - For building installation.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6930

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SPECIFICATIONS

FEDERAL

GG-D-236	- Depth Perception Apparatus, Ophthalmological.
GG-P-455	- Plates and Foils, Photographic (Photosentive Anodized Aluminum).
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

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MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
MIL-V-173	- Varnish, Moisture- and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-B-13207	- Body, Van; Vehicle-Mounted, General Specification for.
MIL-B-13207/3	- Body Van; Vehicle-Mounted, ISO Size 30.
MIL-B-26195	- Box, Wood-Cleated, Skid.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking for US Military Property.
MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

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(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

C73.11 - Dimensions of Plugs and Receptacles-General Purposes, 125 Volts, 15 Amperes, 2 Pole, 3 Wire Grounding Type.
Z35.1 - Accident Prevention Tags, Specification for.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN TRUCKING ASSOCIATION, INC. (ATA)

The Maintenance Council - ATA

Recommended Maintenance Practices Manual.

(Application for copies should be addressed to the American Trucking Association, Inc., The Maintenance Council - ATA, 2200 Mill Road, Alexandria, VA 22314.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT - RULES and REGULATIONS

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 E. Flair Drive, El Monte, CA. 91731.)

INDUSTRIAL FASTENERS INSTITUTE (IFI)

IFI Standards.

(Applications for copies should be addressed to the Industrial Fasteners Institute, 1717 East 9th Street, Cleveland, OH 44114.)

NATIONAL ASSOCIATION OF NAMEPLATE MANUFACTURERS, INC. (NAME)

Nameplate Industry Standards and Practices.

(Application for copies should be addressed to the National Association of Nameplate Manufacturers, Inc. 1700 East Dyer Road, Suite 165, Santa Ana, CA 92705.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02110.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J370 - Bolt and Capscrew Sizes for use in Construction and Industrial Machinery.
J925 - Minimum Access Dimensions for Construction and Industrial Machinery.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The portable and simulator testing and training devices (hereinafter called "devices") shall consist of electromechanical units for testing and training drivers of government vehicles. The type P device (see 3.10) is a portable unit used to test personnel who possess a valid civilian driver's license and are required to operate military vehicles. The type S device (see 3.11) is a simulator capable of in-depth testing of personnel, who may or may not possess a valid civilian drivers's license, in the operation of the type of vehicle(s) required in the performance of their duties. Training device screw-threads shall be in accordance with FED-STD-H28.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of devices from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish devices conforming to this specification.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

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3.3.1 Material deterioration prevention and control. The device shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the device may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the device may be newly fabricated from recovered materials to the maximum extent practicable, provided the device produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the device.

3.4 Human factors engineering. The device shall conform to the human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to, general requirements (4), control/display integration (5.1), visual displays (5.2), audio displays (5.3), controls (5.4), labeling (5.5), anthropometry (5.6), ground workspace design requirements (5.7), environment (5.8), design for maintainability (5.9), hazards and safety (5.13), and personnel-computer interface (5.13) of MIL-STD-1472, as applicable.

3.5 Safety. The devices shall be constructed so as to result in material and components having a correct safety factor, including elimination of nonfunctional sharp edges, protruding points, and excessive lengths of fasteners. Electrical equipment shall be in accordance with the National Electrical Code and shall be effectively grounded to protect all persons and objects from electrical shock hazards.

3.6 Ease of maintenance. Components of the device shall be accessible for servicing, repair, and replacement. The adjustment and replacement of components shall be accomplished with minimum disturbance to adjacent components. Ease of maintenance provisions shall incorporate features insuring operating clearances in accordance with SAE J925, and facilitating maintenance and service operations. Threaded parts shall be in accordance with SAE J370 and IFI Standards.

3.7 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.7.1), with 1 percent copper 8-quinolinolate (by weight), based on the nonvolatile content of the varnish.

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a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.7.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material use complies with Rule 442 South Coast Air Quality Management District.

3.8 Data plates. Data plates shall be furnished and secured to each end item with metal fasteners in accordance with NAME Standards and Practices.

3.8.1 Identification. Identification shall be in accordance with MIL-STD-130 on plates conforming to GG-P-455 and MIL-P-514, type I, style 1, of composition C, type A, class 1 material.

3.8.2 Instruction. Each major component of the device shall be equipped with instruction plates detailing applicable (special) operational, safety, and service procedure to be followed. The safety, warning, caution, and notice plates shall be in accordance with ANSI Z35.1 and shall be compatible with the space available.

3.8.3 Wiring diagram. Plates depicting wiring diagrams shall be furnished on each device. The wiring diagram plate furnished with the type P device shall be mounted on the inside of the cabinet in a location that is visible during maintenance operations. Each major component of the type S device shall be equipped with a simple interface wiring diagram plate, enabling personnel to trace the major interconnecting circuits.

3.9 Interchangeability. All components having the same part number shall be functionally and dimensionally interchangeable. Interchangeable components are defined as two or more like components possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the components themselves or of adjoining components, except for adjustment, and without selection for fit or performance.

3.10 Type P device.

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3.10.1 Design. The type P testing device shall be a portable self-contained unit consisting of the components specified herein for testing of drivers' visual acuity, field of vision, foot reaction, color perception, and depth perception. Typical components and arrangements thereof are shown in figure 1.

3.10.2 Components.

3.10.2.1 Cabinet. The cabinet shall be wood, metal, or reinforced plastic, as specified (see 6.2). The cabinet shall be not more than 24 inches high, 24 inches long, and 18 inches wide. The cabinet shall be equipped with a luggage-type, heavy-duty, carrying handle on top and 4-each wear-resistant feet on the bottom. All components shall be housed and secured within the cabinet when the doors and openings are closed. Clamps, clips, and elastomer foam integral with the cabinet shall be used to prevent component movement. Access doors and openings shall be equipped with tension-type hook latches not less than 2 inches long, with strikers and piano-type hinges. When closed for transport the cabinet shall be weatherproof from rain and dust. The cabinet shall withstand an imposed load of a least 200 pounds on any one square foot of cabinet area, at any one time, without damage or permanent deformation.

3.10.2.2 Visual acuity charts. Unless an optional device is specified (see 6.2 and 6.7), two Snellen charts with a variation in symbol sequence, a variation of the international road signs, and a light source shall be provided. The light shall provide uniform illumination of the charts and shall be not less than 10-foot-candle power. A rack to hold the charts for viewing and a pocket for storage of the charts when not in use shall be provided.

3.10.2.3 Field of vision device. The device for testing an examinee's field of vision shall be metal, wood, or reinforced plastic with the scales of degrees marked on corrosion-resistant metal. The device shall provide for measurement of the lateral range of vision between 40 and 100 degrees on each side of the focus line in units of not more than 5 degrees. Control of the moving targets shall be manual providing selection by the examiner or the examinee of either or both targets. The targets shall be not less than 3/16 inch wide and shall extend to not more than 1 inch above the surface of the base. The targets shall be spring loaded to provide for return of the target upon release of the control to a position not less than the 100-degree mark on each side. A means of mounting or holding the device shall be provided. The field of vision device shall be contained and secured within the folded cabinet during transport. One spare target return spring shall be provided in a labeled storage holder in the cabinet.

3.10.2.4 Depth perception. The device for measuring an examinee's depth perception shall be provided with one of two optional methods:

a. Opthalmological appartus per GG-D-236, type I, modified for use within the cabinet. A scale for measuring the degree of an examinee's depth perception shall be provided.
b. Three targets are located on slides in pre-selected set positions within the tester cabinet. A dial on the cabinet shall enable the examiner to offer the targets in any one of four or more different combinations of

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positions. The examinee is asked to identify which target appears nearest and farthest from his position of view. Multiple condition responses can be recorded to establish an acceptable level of depth perception.

The depth perception device shall be uniformly illuminated by a 10 foot-candle power light source.

3.10.2.5 Electrical cable. The 110-volt (V), ac, 3-connector electrical power supply cable for connection of the testing device to a power source, shall not be less than 10 feet long, and shall be in accordance with NFPA 70, type SJT. Strain relief for the cable shall be provided inside the cabinet.

3.10.2.5.1 Connector plug. Unless otherwise specified (see 6.2), the cable shall be provided with a 125V, 15 amp, 3-connector plug conforming to ANSI C73.11, and shall be approved for the purpose intended by a nationally recoginized testing agency such as Underwriters' Laboratories, Incorporated, indicating conformance to all applicable requirements of the agency. Approval shall be evidenced by the attachment of the agency seal, label, or listing mark indicating the standard or specification under which the connector plug was approved.

3.10.2.6 Indicator lights. A minimum of 2 indicator lights shall be provided. The indicator lights shall be red and green and shall operate as functional components of the foot-reaction device, with independent illumination control. The lights shall be round-frame type, grouped together, and mounted in the timer box portion of the cabinet. The lights shall be installed so that they do not project more than 1/4 inch beyond the flat surface of the cabinet. The lights shall be visible to the examinee.

3.10.2.6.1 Lenses. Lenses shall be interchangeable, colored, faceted button type. A spare lens of each color and a storage place for each lens shall be provided. The lenses shall be mounted with a clearance of not less than 1 inch between the removeable rings to allow for easy replacement of lenses and bulbs.

3.10.2.6.2 Bulbs. Bulbs shall be pin-bayonet type for operation on 110 to 120V, 60 Hz.

3.10.2.7 Timer. The timer shall be an electrical unit operating on 110 to 12V, 60 Hz. The timer dial shall be divided into 100 units of 1/100 second, with each tenth division numbered. The indicating pointer shall extend to within 1/8 inch of the dial markings. A method of manually resetting the pointer to zero shall be provided. The timer shall operate with an accuracy of 0.03 seconds.

3.10.2.7.1 Timer control button. The red and green indicator lights shall be operable by the examiner through the use of the timer control button which shall be located out of visual range of the examinee. Depression of the timer control button by the examiner shall simultaneously extinguish the green light, illuminate the red light, and start the timer.

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3.10.2.8 Control cord. The electrical control cord connecting the timer to the foot panel shall be in accordance with NFPA, type SO, and shall be not less than 6 feet long, flexible, minimum size 18 AWG.

3.10.2.9 Foot panel. The foot panel shall be of wood or corrosion-resistant metal. The bottom surface of the foot panel shall be equipped with non-slip material to prevent movement on the floor. Two pedals, one serving as a brake and the other as an accelerator, shall be operated by the right foot of the examinee. A left-foot rest shall be provided and shall be covered with corrosion-resistant metal.

3.10.2.9.1 Accelerator pedal. The accelerator pedal shall require a continuous pressure for illumination of the green light; removal of pressure from the accelerator or illumination of the red light shall extinguish the green light.

3.10.2.9.2 Brake pedal. The brake pedal shall be spring loaded. Depression of the brake pedal 1/2 inch or more shall stop the pointer on the timer and activate the red light. Releasing the pedal extinguishes the red light.

3.10.2.10 Color perception device. The color perception device shall be of the pseudo-isochromatic-color-plate type. A minimum of 14 different plates having digits providing various color discrimination shall be furnished. The plates and the holder for the booklet shall be stored and secured within the cabinet during transport.

3.10.2.11 Optional glare recovery night blindness device. When specified (see 6.2), the device for measuring glare recovery night blindness shall be made of two standard headlight sealed beams mounted in the testers, along with a tenth of a second timer, labeled switches for selecting one of several number or letter sequences displayed on the cabinet. Sealed beams are to be viewed for 5-10 seconds. Examiner switches off headlights which automatically starts tenth of a second timer and displays a pre-selected number or letter sequence on the cabinet surface. The examiner switches off timer when he hears the examinee correctly relate the displayed number or letter sequence. Number or letter display to be switched off seperately. Timer has recorded examinees' delay in visual recovery.

3.10.3 Visibility.

3.10.3.1 Glare. The characteristics of the depth perception and visual acuity components shall provide for operation by the examinee without direct light or glare from the source of illumination. The angle of incidence and refraction shall not be less than 10 degrees.

3.10.3.2 Obstruction of vision. The characteristics of the field of vision component shall provide for operation by the examinee without obstruction of vision.

3.10.4 Treatment and painting. The portions of the type P device normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. The exterior of the device shall be painted green, Color No. 24084.

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The interior portion containing the depth perception device shall be black, Color No. 37038, except that the back section opposite the examinee viewing window shall be white, Color No. 37875.

3.11 Type S device.

3.11.1 Design. The type S, automated driver-training testing device shall function to provide a driver with tactile, audio, and visual simulation of vehicle driving. Figure 2 shows a typical simulator van installation. The type S device shall provide a simulation of the vehicle cab, including:

a. Seating, instruments, and controls with simulator feedback to driver and instructor.
b. Audio signals in response to driver input.
c. Visual site or roadway interface in response to driver input.

3.11.1.1 Components. The components shall include a mock-up of the driver's station of the intended vehicle, an instructor's console, the sound simulation system, the computer control system, and the audio-visual system. The driver's station shall be a full-scale replica and include all operating controls and indicators necessary for accurate simulation. Commercially available parts shall be utilized in the driver's station to provide both realism and readily available replacements. The instructor's control console shall house controls and indicators utilized to originate situations, insert simulated vehicle malfunctions or problems, and monitor the trainee. The sound simulation system shall electronically generate the sounds associated with vechile operation. The volume of the system shall be under the control of the instructor. Multiple speakers shall be furnished. The computer control system shall be of the solid-state type utilizing integrated circuits and removable circuit boards to the maximum extent practicable. The computer shall correlate the instructor's input with the driver's operation for the various situations, based on vehicle loadings and speeds, and provide feedback to the driver. The audio-visual system shall add interactive audio and visual stimuli automatically to the driver and provide override control from the instructor. The visual portion shall include a changing scene of the operation site or the roadway.

3.11.1.2 Installation. The type S, class 1 device shall be furnished and installed in a van body conforming to MIL-B-13207. Unless otherwise specified (see 6.2), the van body features of MIL-B-13207/3 shall be as follows:

a. Size 30 van body (ISO designation 1B container).
b. Semitrailer mounted (without dolly converter).
c. Noise-absorption material required in partition separating simulator and classroom, as a minumum.
d. Windows shall be required in classroom portion of van bodies only, however, blackout curtains are not required.
e. Contractor installation of type S device is required.
f. Utility electrical system components in the simulator room shall be as required by the device.

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The type S, class 2 device shall be arranged for installation within a building with dimensions and details as specified (see 6.2). Complete instructions for installation shall be furnished by the contractor.

3.11.2 Requirements characteristics. The type S device detailed requirements shall be as specified (see 6.2 and 6.6). The type S device video and audio training aids shall be in accordance with RP-503 from the Recommended Maintenance Practices Manual of the American Trucking Association.

3.11.3 Environmental.

3.11.3.1 Storage temperatures. The type S device shall not be damaged by storage in any ambient temperature from minus 40 °F (minus 40 °C) to plus 120 °F (plus 49 °C).

3.11.4 Treatment and painting. The type S device shall be treated and painted to prevent corrosion, in accordance with the contractor's standard practice. For the class 1 device, the van body shall be painted in accordance with MIL-B-13207, except that the portion housing the simulator shall be finish coated in accordance with the simulator contractor's recommendations.

3.12 Workmanship. All workmanship of each driver testing and training device shall be in accordance with the engineering, manufacturing, and production standards of the driver-training device industry and shall, in addition, be in accordance with MIL-STD-454, requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. The first article type P device shall be tested as specified in 4.5.2.1. The first article type S device shall be tested as specified in 4.5.2.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.1.1 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.0 percent defective.

4.4.2 Tests.

4.4.2.1 Sampling. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The devices shall be examined as specified herein for the following defects:

Major

101. Materials not as specified (see 3.3).
102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.3.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
104. Contractor does not have documentation available for identification of material, material finishes, or treatment (see 3.3.3).
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the device (see 3.3.4).
106. Safety provisions not as specified (see 3.5).
107. Ease of maintenance provisions not as specified (see 3.6).
108. Fungus and moisture resistance treatment not as specified (see 3.7).
109. Data plate not as specified (see 3.8).

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110. Interchangeability not as specified (see 3.9).
111. Design not as specified (see 3.10 type P and 3.11 type S).
112. Components missing or not as specified (see 3.10.2 or 3.11.1.1). Each component missing or not as specified shall constitute one defect.
113. Human factors engineering not as specified (see 3.4).
114. Environmental conditions for type S device not as specified (see 3.11.3.1).
115. Treatment and painting not as specified (see 3.10.4 or 3.11.4).
116. Workmanship not as specified (see 3.12).

4.5.2 Tests.

4.5.2.1 Type P tests. Connect the device to a 110 - 120V, 60 Hz power source and perform functional tests of components as follows:

a. Position the complete testing and training device and cabinet on a level, concrete surface. Impose a pressure load of not less than 200 pounds on the side of the cabinet and within an area of one square foot. Repeat in two other areas on the same side including the middle. Repeat on all other sides. There shall be no evidence of permanent deformation or damage, either outside or inside.
b. Energize the Snellen chart and depth perception illumination light. With a light meter, measure the illumination at the center of the chart. Illumination at the center of the chart shall be not less than 10 foot-candle power.
c. Operate the field of vision device mechanism. It shall operate smoothly and targets shall traverse to 100- to 40-degree range. When the mechanism is released, the appropriate target shall return to rest at not less the 100-degree mark.
d. Observe depth perception rods through the viewing door and operate mechamism. Mechanism shall operate smoothly without binding or malfunction. For target multi-position selection option, operate dial knob to each position and insure that each test condition registers a different spacing of targets from the viewing face of the cabinet.
e. In sequence, depress the control button for the red and green lights. Corresponding light shall illuminate as each control button is depressed.
f. Verify that each of fourteen psuedo-isochromatic plates are different for selection in color blindness testing.
g. Depress control button of foot reaction timer device. Depressing the control button shall simultaneously extinguish the green light, illuminate the red light, and start the timer.
h. Depress accelerator pedal; green lamp shall illuminate. Remove foot from accelerator pedal, then depress again. Green light shall extinguish when pressure is removed and shall illuminate when pressure is reapplied.
i. Depress the timer control button extinguishing the green light, illuminating the red light, and starting the timer. Depress the brake pedal; the timer shall stop and the red light shall remain illuminated. Release timer control button extinguishing red light, and remove foot from the brake pedal. Reset the timer pointer by manually advancing the pointer to zero.

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j. Connect an electric stopwatch, calibrated to 1/10 of a second accuracy, in parallel with the red indicator light of the timer. Start the timer and stopwatch simultaneously by depressing the control button. Allow the timer and stopwatch to run for a minumum of 90 seconds and release the control button, stopping the timer and stopwatch simultaneously. The timer shall operate with an accuracy of 0.03 seconds.
k. Switch on headlights. Switching headlights off should automatically start tenth of a second timer and turn on the light(s) for display of a number or letter sequence. Test operation of number sequence display selection dial and timer stopwatch.

Inability of any component to function as required in section 3 shall constitute failure of this test.

4.5.2.2 Type S tests.

4.5.2.2.1 Performance. Using the simulator manufacturer's installation and operation instructions furnished with type S device, set up and connect all components. Verify correct voltage, amperage, decibel level, and light lumins. Using an inexperienced operator-driver, perform training operations for not less than five 8-hour periods, using all visual and audio effects available. At the conclusion of the test, verify correct physical operation and verify correct technical specification tolerances of components. Any function, operation, or technical value not in conformance with the manufacturer's performance standards or tolerances, permanent deformation, nonconformance with any requirement specified herein shall constitute failure of this test.

4.5.2.2.2 Storage temperature. The contractor shall, in lieu of testing, submit evidence acceptable to the contracting officer prior to approval of the first article device that the type S device(s) proposed to be furnished under this specification meets the requirements of 3.11.3.1.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Examination. Each training device prepared for shipment shall be examined for the following applicable defects. Any defects detected shall be corrected prior to shipment of the device.

117. Components of type P devices not secured in cabinet for levels A or C.
118. Type P device not preserved as specified for level A.
119. Type S, class 1 devices not preserved and secured in van as specified for levels A or C.
120. Van chassis for type S, class 1 devices not preserved as specified for level A.
121. Type S, class 2 devices not preserved and secured to container base as specified for level A or C.

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122. Containers not as specified for type P and type S, class 2 devices for levels A or C.
123. Strapping not as specified for level A.
124. Marking illigible, incorrect, incomplete, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Type P testing devices. The components of each type P testing device shall be secured within the cabinet provided. Cushioning material in accordance with MIL-STD-1186 shall be used to pad all sharp corners and projections of components to prevent damage from abrasion. Technical publications shall be placed in the cabinet. Each complete testing device shall then be preserved in accordance with MIL-P-116, method IIb. The cabinet shall serve as the inner container, and the outer container shall be a box conforming to PPP-B-636, V11c, style optional.

5.1.1.2 Type S testing and training devices.

5.1.1.2.1 Class 1 devices. All components shall be secured in the transport position. Technical publications shall be preserved in accordance with MIL-P-116, method IC-3. Components or accessories not provided with transport storage facilities shall be preserved in close-fitting boxes conforming to PPP-B-636, class weather resistant, style optional. The van chassis shall be preserved in accordance with the level A provisions of MIL-STD-281 for mobile shipment.

5.1.1.2.2 Class 2 devices. All electric and electronic-controlled components shall be preserved in accordance with MIL-P-116, method IIa or IIb, as applicable. Large components capable of being bolted to the base of the shipping container shall be preserved in accordance with method IIa. All other components shall be preserved in accordance with method IIb. The outer container for components preserved in accordance with method IIb shall be a box conforming to PPP-B-601, domestic type, style optional. Components preserved in accordance with method IIa shall be bolted directly through the barrier to the base of the shipping container specified in 5.2.1.2.2. The components that are not electric or electronic controlled shall be preserved in close-fitting boxes conforming to PPP-B-601, domestic type, style optional. Cushioning shall be provided to prevent damage to components within the containers during shipment and handling. Strapping of the boxes shall not be required. Technical publications shall be preserved in accordance with MIL-P-116, method IC-3.

5.1.2 Level C.

5.1.2.1 Type P devices. All components shall be secured within the cabinet and cushioned as specified for level A. Technical publications shall be preserved with method IC-3, MIL-P-116. The cabinet shall be closed and latched.

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5.1.2.2 Type S devices.

5.1.2.2.1 Class 1. Preservation shall be as specified for level A except that boxes for components not mounted shall be class domestic.

5.1.2.2.2 Class 2. Components having mounting provisions shall be bolted to the base of the shipping container. All other components shall be placed in domestic type containers, either wood or fiberboard as directed by the size and weight of the components. Cushioning shall be used as necessary to immobilize components within the containers.

5.2 Packing. Packing shall be level A or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Type P testing devices. Each type P testing device preserved as specified in 5.1 shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, style optional. Strapping shall conform to QQ-S-781, class 1, style I or IV, finish B, size as applicable.

5.2.1.2 Type S testing and training devices.

5.2.1.2.1 Class 1 devices. The preserved technical publications and boxed components and accessories shall be secured within the van. The van shall then be closed and secured for shipment.

5.2.1.2.2 Class 2 devices. Each complete device shall be packed in a container conforming to MIL-C-104, type I, class 2, style a, or to MIL-B-26195, type II, style A, class 1 whichever is appropriate for the weight of contents. Components preserved in accordance with method IIa shall be bolted directly through the barrier material to the base of the container. Boxed components shall be secured to the container base and braced and blocked to prevent movement in accordance with the appendices to the container specifications and to MIL-STD-1186.

5.2.2 Level C.

5.2.2.1 Type P devices. No additional packing is required. When multiple quantities of devices are to be shipped to a single destination, they may be consolidated on 40 x 48-inch wood pallets.

5.2.2.2 Type S devices. Class 1 devices shall be packed as specified for level A. Complete class 2 devices shall be individually packed in containers suitable for the size and weight of the contents; blocking, bracing, and tie-down shall be applied as necessary to prevent movement and damage to the contents.

5.3 Marking. Marking for shipment and storage shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The type P driver testing device is intended for use in the testing of applicants for military vehicle operator's license. The type S simulator is intended for in-depth training and testing of both new and experienced personnel for vehicle operation in a simulated driving atmosphere of the particular vehicle(s) they will be operating.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class required (see 1.2).
c. When a first article is not required (see 3.2).
d. Time frame required for submission of first article and number of devices required (see 3.2).
e. Type or types of cabinet material required (see 3.10.2.1).
f. When an optional visual acuity devise is required and the minimum specifications of the device (see 3.10.2.2 and 6.7).
g. Type of connector plug required if other than specified (see 3.10.2.5.1).
h. When glare recovery night blindness option is to be supplied, (see minimum specifications, 3.10.2.11 and 4.5.2.1.k).
i. When the van body for type S, class 1 device is required to have options of MIL-B-13207 different than those specified (see 3.11.1.2).
j. Dimensions and details of building, if known (see 3.11.1.2).
k. Size of ISO container required (see 3.11.1.2).
l. Minimum detailed component requirements of type S device (see 3.11.2).
m. Level of preservation and of packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the items should be a first produced unit. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each device.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts, test equipment, and maintenance tools as necessary (including any special tools), and instructions regarding shipment of each device.

6.6 Type S device requirement characteristics. The procuring activity should specify the minimum components, characteristics, and performance of the type S device. These should include specifications on the following.

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a. Type of vehicle operator-driver compartment, i.e., light truck, heavy truck, earth-moving or construction equipment, materials-handling equipment or other type of vehicle(s).
b. Operator instrument and control, visual, feel, and function characteristics, as similar as practical to the vehicle(s) selected in a. above.
c. Type of terrain, road, and visual-audio effects required for driver input, such as earth-moving job site, materials-handling equipment marshalling yard, highway, city streets, pioneer trail, etc.
d. Minimum component coverage, including:

(1) Driver's cab assembley.
(2) Instructor's console assembly.
(3) Audio sound system.
(4) Visual terrain projectors with minimum 180-degree, 270-degree, or 360-degree screen, dependent on field of view from the simulator.
(5) Interconnection electrical cables.
(6) Electric power source characteristics.
(7) Regular production options and special options requirement.

6.7 Optional visual acuity device. When the visual acuity component specified herein for the type P device does not fully meet the requirements of the procuring activity, the procuring activity should consult their local optometry clinic for specifications and assistance.

6.8 Optional glare recovery night blindness. When the glare recovery option is to be offered as integrated into the basic portable test device, the prospective supplier shall provide dual bids, i.e., basic unit cost and unit cost with glare recovery tester.

6.9 Optional motion simulation. When motion to the student is to be simulated, the type of motion shall be determined (e.g., motion platform, seat shaker, etc.)

6.10 Information figures. Figure 1 shows a typical type P testing device which has been found acceptable; however, the figure is included for illustration only and is not intended to preclude the furnishing of another testing device which conforms to this specification. Figure 2 shows ISO van installation for a typical type S device.

6.11 Subject term (key word) listing.

Building installation
Driver testing device
Driver training device
Portable
Simulator
Van mounted

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6.12 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - TD
Air Force - 99

Review activity:
Army - MD

User activities:
Navy - MC, SA
Air Force - 70

Preparing activity:
Army - ME

Project 6930-0178

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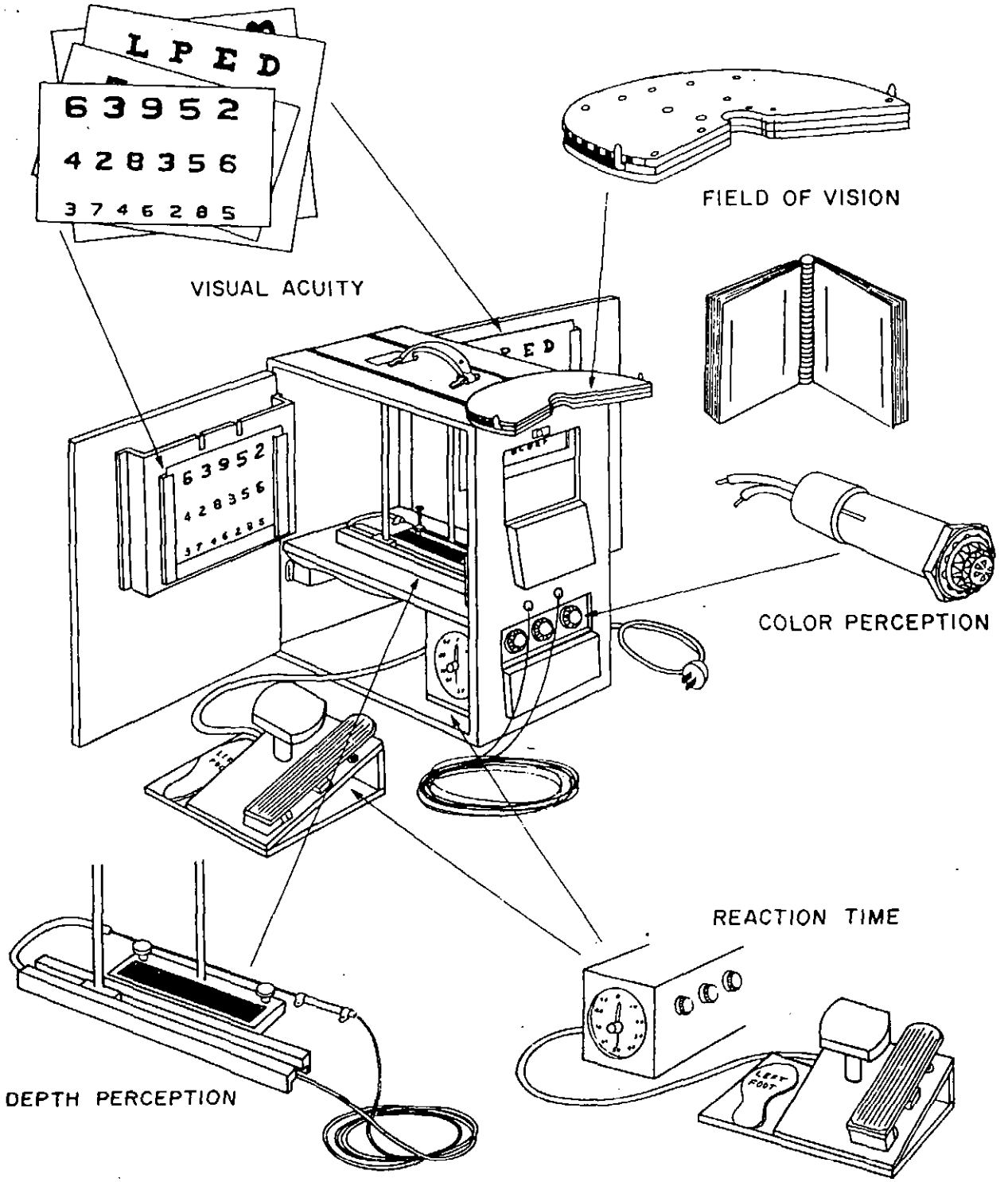


FIGURE 1. Components of one type of driver testing device.

X-3148A

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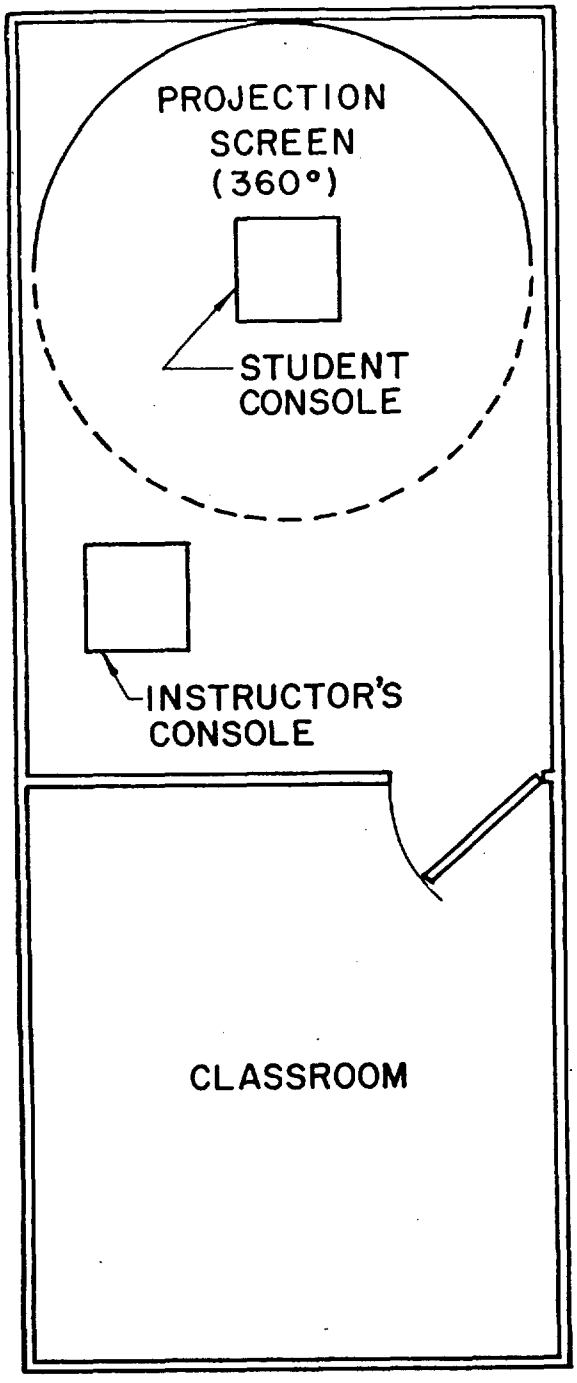


FIGURE 2. TYPICAL TYPE S SIMULATOR
ISO VAN INSTALLATION

X-3149

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-10795F	Driver Testing and Training Device, Portable and Simulator

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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